LECTURE 1 · WEDNESDAY 28 OCTOBER 2026

Meet AI

What AI actually is, the main tools you can use, and how to get access.

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Who am I?

ENGINEER

B.Sc. (Hons) Computer Engineering, University of Malta.

RESEARCHER

Research Support Officer on Project M.AI.ESTRO, Faculty of ICT: AI and virtual reality that guide people learning hands-on skills.

DAILY AI USER

I use AI every day for writing, coding and planning. These slides were built with AI help, and every slide was checked and decided by me.

You do not need to be an engineer for this course. That part is my job.

How this course works

BRING A LAPTOP

Every week. We work hands-on with real tools.

WI-FI: EDUROAM

Log in with your UM IT account. Setup guide: um.edu.mt/itservices/facilities/eduroam

ATTENDANCE

It is the only assessment. Eight Wednesdays, 12:00 to 14:00.

AFTER EACH LECTURE

Slides and revision notes go on the VLE. Short break at 13:00.

28 Oct	**Meet AI**: what AI is, the main tools and how to get access.
04 Nov	**Tokens and better results**: how AI reads your words, and how to ask well.
11 Nov	**Research and documents**: search, summarise, compare and check.
18 Nov	**Writing and communication**: emails, reports and messages for any audience.
25 Nov	**Planning, meetings and your own assistant**: plans, minutes, and building a Gem.
02 Dec	**Presentations and visuals**: structure, slides, speaking notes and images.
09 Dec	**Tables and data**: ask questions of a spreadsheet, no coding.
16 Dec	**Your AI workflow**: a repeatable process for your own field.

By December, you will be able to…

1
Pick the right type of AI tool for a task.

2
Write clear instructions that get useful answers.

3
Use AI for research, writing, planning, presentations and data.

4
Improve answers instead of accepting the first one.

5
Build reusable prompts and workflows for your field.

6
Know when to check, ask an expert, or decide yourself.

BEFORE WE START

Who's in the room?

Hands up, and I'll count. There are no wrong answers.

CLASS SURVEY

How often do you use AI tools?

Never	0
Once or twice	0
Every week	0
Every day	0

CLASS SURVEY

Which have you used? Hands up for each one.

ChatGPT	0
Gemini	0
Claude	0
Microsoft Copilot	0
Something else	0

CLASS SURVEY

Do you pay for an AI tool?

Free only	0
Paid plan	0
Through work or uni	0
Not sure	0

CLASS SURVEY

What have you used it for?

Study	0
Work	0
Everyday life	0
Not yet	0

CLASS SURVEY

Which area is your degree closest to?

Business	0
Health and care	0
Justice and social work	0
Design and arts	0
Science and education	0

CLASS SURVEY

How do you feel about AI?

Excited	0
Curious	0
Sceptical	0
Worried	0

PART A

What AI actually is

No maths and no coding. Just a clear picture of what is going on inside.

Three common myths

MYTH

“It knows everything.”

It learned patterns from text. There is no list of facts inside to look up.

MYTH

“It searches the internet for the answer.”

A chatbot writes from what it learned. Some tools can also search, but that is an extra feature.

MYTH

“It thinks like a person.”

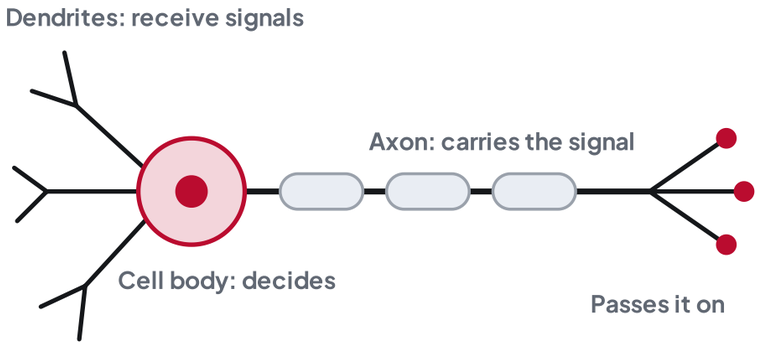
It predicts likely words. Very useful, but different from thinking.

IN ONE SENTENCE

AI chatbots learn patterns from huge amounts of text, then predict what comes next.

Hold on to this sentence. Everything today builds on it.

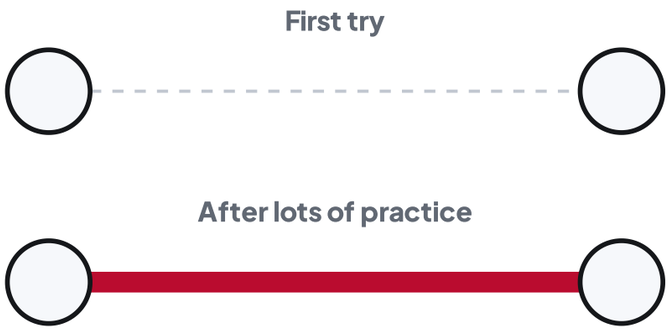
Start with your brain



- About **86 billion** neurons.
- Each one connects to **thousands** of others.
- A neuron collects signals. If they are strong enough, it **fires** and passes a signal on.

Neuron count: Azevedo et al. (2009), *Journal of Comparative Neurology*.

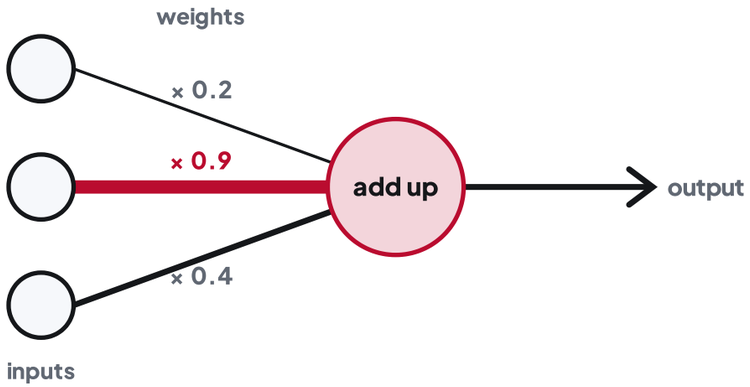
Learning = stronger connections



- Practise something, and the connections you use get **stronger**.
- That is why the hundredth bike ride is easier than the first.
- Neuroscientists sum it up as: **neurons that fire together, wire together**.

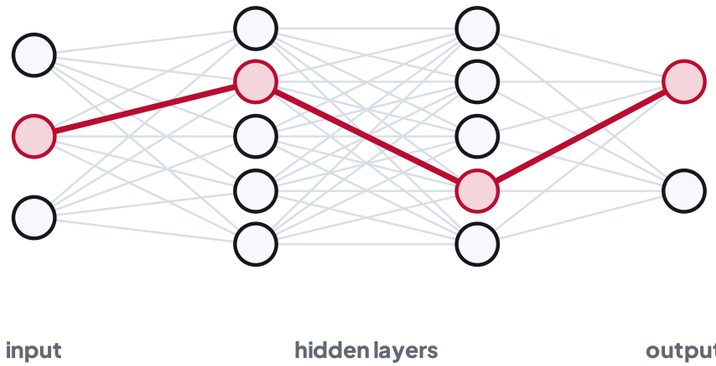
Idea from Donald Hebb, *The Organization of Behavior* (1949).

An artificial neuron: a simple maths copy



- It takes in numbers.
- Each input has a **weight**: how much it matters.
- It adds them up and passes a result on.
- The weights play the role of **connection strength**.

Connect lots of them: a neural network



- Layers of artificial neurons pass numbers forward.
- Big models have **hundreds of billions** of weights. Meta's open model Llama 3.1 has 405 billion.
- Your brain has roughly **100 trillion** connections: still far more.

Meta, *The Llama 3 Herd of Models* (2024). Synapse count is a common estimate; figures vary.

How it learns: guess, check, adjust

1
Show it some text with the next word hidden.

2
It guesses the missing word.

3
Compare the guess with the real word.

4
Nudge the weights so the right word becomes a little more likely.

Repeat this trillions of times. Nobody types in the rules: the patterns come from the examples.

A bit like a child learning a language

A CHILD

- Hears thousands of sentences.
- Nobody teaches grammar rules first.
- Starts to predict what people will say.

AN AI MODEL

- Reads trillions of words.
- Nobody programs the rules.
- Learns which words tend to follow which.

One big difference: a child also sees, touches and lives in the world. The model only has text.

GAME

Finish the phrase

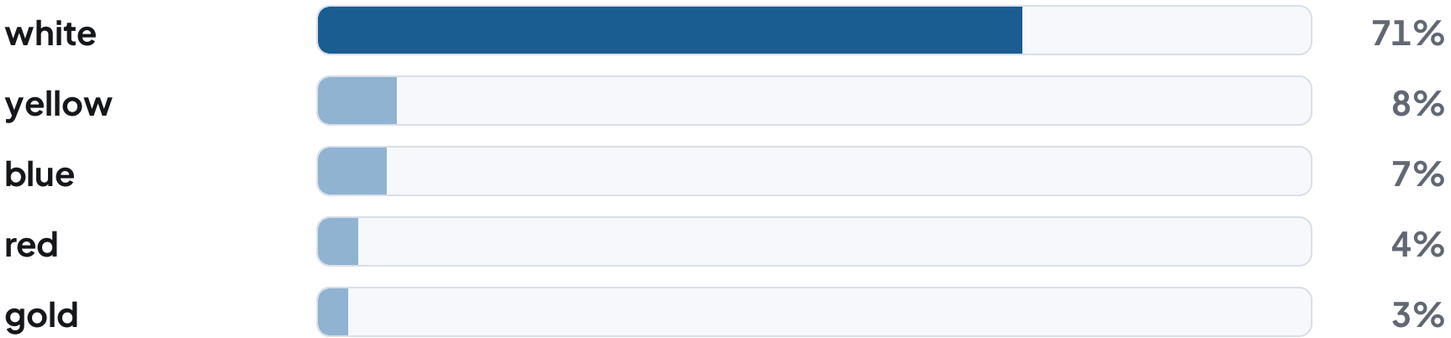
Shout out the first word that comes to mind.

FINISH THE PHRASE

Black and _____

WHAT A MODEL MIGHT PREDICT

Black and …



Illustrative numbers, to show the idea

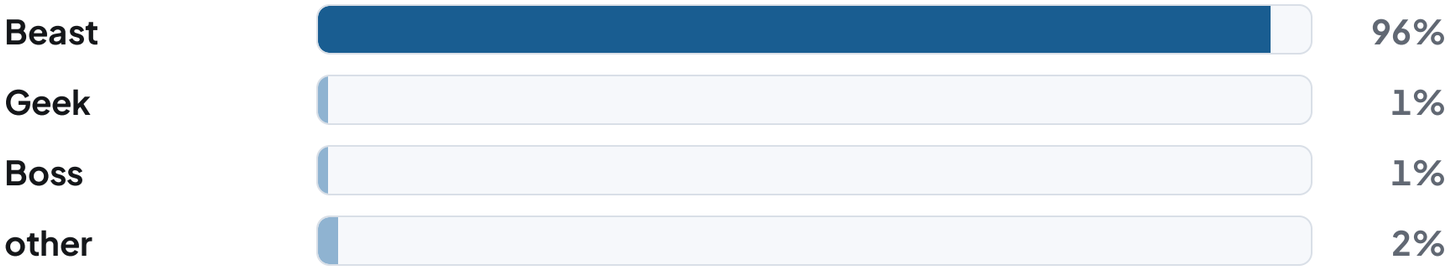
Most of you said **white**. Your brain picked the most familiar continuation. That is a prediction.

FINISH THE PHRASE

Beauty and the _____

WHAT A MODEL MIGHT PREDICT

Beauty and the …



Illustrative numbers, to show the idea

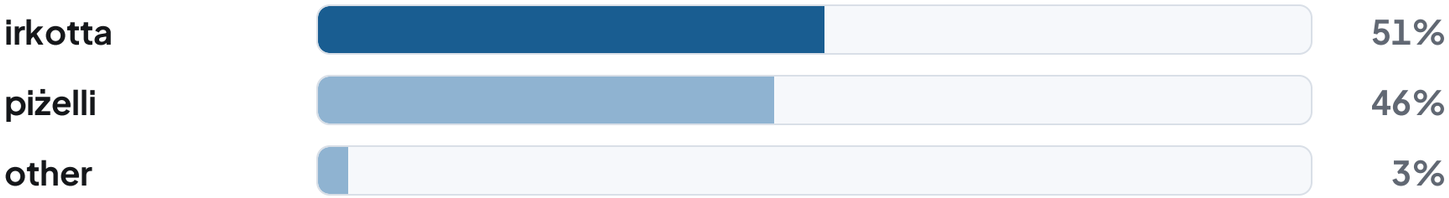
When one continuation dominates everything we have heard, the prediction is almost certain.

FINISH THE PHRASE

Pastizzi tal-_____

WHAT A MODEL MIGHT PREDICT

Pastizzi tal- …



Illustrative numbers, to show the idea

Two answers are almost tied. The model picks one, and if you ask again it may pick the other. That is why AI can give **different answers to the same question**.

FINISH THE PHRASE

The capital of Malta is

WHAT A MODEL MIGHT PREDICT

The capital of Malta is …



Illustrative numbers, to show the idea

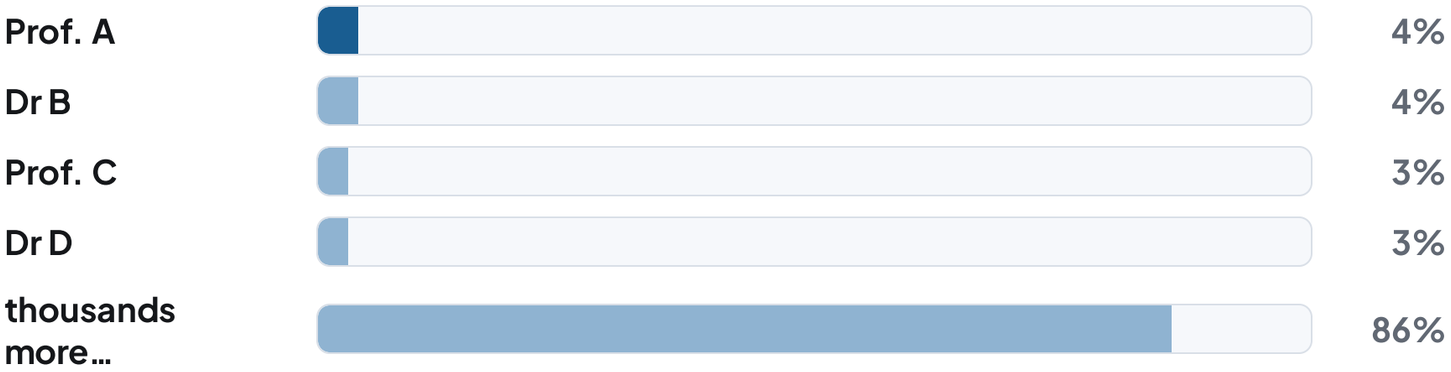
Facts that appear thousands of times come out right. But they are still **predictions**, not looked up.

FINISH THE PHRASE

The best lecturer at the University of Malta is

WHAT A MODEL MIGHT PREDICT

The best lecturer at the University of Malta is …



Illustrative numbers, to show the idea

There is no real answer, but the model will still write a fluent, confident one. This is where **made-up answers** come from.

What the game shows

YOUR BRAIN PREDICTS TOO

You finished every phrase without thinking about it.

AI DOES IT AT SCALE

One piece of a word at a time, again and again, to write whole answers.

LIKELY IS NOT TRUE

Usually right for common facts. Confidently wrong for rare ones.

SCALE

Meta's Llama 3 learned from about 15 trillion pieces of text.

Reading non-stop at 250 words a minute, 24 hours a day, a person would need roughly **85,000 years** to read that much.

Meta AI, Llama 3 announcement (April 2024): over 15 trillion tokens. Reading time is a rough estimate at about 0.75 words per token.

Why it sounds so sure of itself

WHAT IT IS TRAINED TO DO

Produce text that is **likely** and **sounds right**.

WHAT IT IS NOT BUILT TO DO

Know whether a statement is **true**.

So it is excellent at drafting, explaining and rewording. It still needs you to check facts, numbers and sources.

One more thing: it does not read words

AI splits text into **tokens**. Common words are one piece; rarer words are cut into chunks.

ENGLISH: 9 TOKENS

`Un` `bel` `ievable` ` results` ` at` ` the` ` University` ` of` ` Malta`

MALTESE: 2 WORDS, 6 TOKENS

`Past` `izzi` ` tal` `-` `irk` `otta`

Split with OpenAI's o200k tokenizer, used by recent ChatGPT models.

Next week: how tokens explain cost, usage limits, and why very long chats go wrong.

Part A in three sentences

1

AI chatbots are trained on huge amounts of text to predict what comes next.

2

What they learned is stored as billions of weights, not as a database of facts.

3

Likely is not the same as true. You stay the checker and the decision-maker.

BREAK

5 min

Back at 13:05. Next: the big three AI companies, their tools, and how to get access.

PART B

The tools, and how to get them

The big three companies, what each tool is good at, how to stay safe, and how to get access for free.

The big three

OPENAI → CHATGPT

Started the chatbot boom in November 2022. The most widely used AI assistant.

GOOGLE → GEMINI

Built into Google's apps. **Your UM account already includes it.**

ANTHROPIC → CLAUDE

Founded in 2021 by former OpenAI researchers. Known for careful writing and coding.

All three are large language models trained the way you saw in Part A. They differ in style, features and privacy settings.

Plenty of other names, often the same engines

Microsoft Copilot

Built into Windows and Office. Runs on OpenAI models.

Meta AI

Inside WhatsApp, Instagram and Facebook.

Perplexity

An AI search engine that shows its sources.

Mistral

A European (French) AI company.

DeepSeek

A Chinese model that is free to use.

Thousands of apps

Many “AI apps” are a nice screen on top of one of the big models.

Free, paid, or free through someone else

FREE PLANS

All three have one. Usage limits, but **enough for this course**.

PAID PLANS

Roughly €5 to €25 a month: higher limits, the strongest models, extra features.

PAID FEATURES FOR FREE

Gemini through your **UM account**. ChatGPT Plus through **AI for All**. ChatGPT Go through some **Revolut** plans.

You do not need to pay for anything in this course.

Prices from the official pricing pages, October 2026. They change often.

LIVE DEMO

Same question, three assistants

```
I am a first-year student at the University of Malta.
Give me 5 short tips on how I could use AI to prepare for an exam,
and 1 thing I should be careful about.
```

Watch for: length, structure, tone, and anything that sounds wrong.

LIVE DEMO

Which answer did you prefer?

ChatGPT		0
Gemini		0
Claude		0
No real difference		0

WHAT THE DEMO SHOWS

There is no single "AI answer".

Compare, combine the best parts, and then **you** decide.

Two kinds of AI tool

CHAT ASSISTANT

- You type, it replies.
- Works in a browser tab or an app.
- Only sees what you give it.
- ChatGPT, Gemini, Claude.

AGENT

- You give it a goal; it works through many steps on its own.
- Can open and edit files, run programs, use websites.
- Claude Code, OpenAI Codex, ChatGPT agent mode.

What they look like

Chat assistant

Can you make my email to my lecturer sound more polite?

Of course. Here is a more polite version: “Dear Dr Borg, I hope you are well…”

Shorter please.

Agent working on your computer

```
> Tidy up my thesis folder and fix the references
Reading 14 files…
✓ Renamed 6 chapter files
✓ Fixed 23 references in chapter3.docx
? About to delete old_draft.docx. Allow?
```

Illustrations, not real screenshots.

In this course we use chat assistants

- They are free, safe to learn with, and do not touch your files.
- Everything you learn about asking well also works with agents later.
- We come back to agents and automation in Lecture 8.

What can AI tools do for you?

Chat and writing

Explain, draft, rewrite, brainstorm.

Search and research

Find sources with links: Gemini Deep Research, Perplexity.

Your documents

Ask questions about your own PDFs and notes: Gemini Notebook (NotebookLM).

Images and presentations

Visual ideas, slide outlines, illustrations.

Meetings and audio

Transcripts, minutes, action lists, audio overviews.

Tables and data

Ask questions of a spreadsheet, spot patterns, make charts.

QUICK QUIZ

Which type of tool would you use?

1

Summarise a 40-page reading for tomorrow's seminar

2

Find recent research on burnout among nurses

3

Turn rough notes into a polite email to your placement supervisor

ANSWERS

Which type of tool would you use?

Your documents

Summarise a 40-page reading for tomorrow's seminar.

Upload it to Gemini Notebook and ask questions about it. It answers from your file, not from memory.

Search and research

Find recent research on burnout among nurses.

A plain chatbot may invent papers. Use a tool that searches and shows links, then open the sources.

Chat and writing

Turn rough notes into a polite email to your placement supervisor.

Any chat assistant. Give it the notes, the audience and the tone you want.

USING AI SAFELY

Stay in charge

When to check, what not to share, and the settings that protect your data.

Check it, ask an expert, or decide yourself

CHECK IT

Facts, numbers, dates, quotes and references. Anything you will repeat to others.

ASK AN EXPERT

Health, legal, financial or safety questions. Anything with real consequences for a real person.

DECIDE YOURSELF

Judgement calls, decisions about people, and anything you will put your name to.

What not to paste into an AI tool

- **Other people's personal details**: patients, clients, service users, classmates.
- **Passwords**, ID numbers and bank details.
- **Confidential work documents** you are not allowed to share.
- **Unpublished research or exam material** that is not yours to share.

A simple test: if you would not pin it on a public noticeboard, think twice.

AI in assessed work

- Rules differ between units and lecturers. **Always check your assignment instructions.**
- If AI use is allowed, be open about how you used it.
- Keep a record: save your prompts and the AI's answers.
- You are responsible for everything you submit, including AI mistakes.

See the University of Malta guidance document "Using Generative AI in student work", available on the VLE.

Your data, your settings

TOOL	STOP TRAINING ON YOUR CHATS	PRIVATE ONE-OFF CHAT
ChatGPT	Settings → **Data controls** → turn off **Improve the model for everyone**	**Temporary chat**
Gemini (personal Google)	Gemini Apps Activity → turn off **Keep Activity**	**Temporary chat**
Gemini (UM account)	Nothing to do: not used for training and not reviewed by humans	n/a
Claude	Settings → **Privacy** → turn off model improvement	**Incognito chat**

Sources: OpenAI Data Controls FAQ; Google Gemini Apps Privacy Hub; Anthropic Privacy Center; UM IT Services Google Workspace page (October 2026).

UNIVERSITY OF MALTA IT SERVICES

With your UM account, Gemini does not train on your chats and no human reviews them.

That is why your UM account is the right place for anything to do with your studies.

um.edu.mt/itservices/google

GETTING ACCESS

Your setup for this week

Two essentials: Gemini with your UM account, and a ChatGPT account. Then two optional extras.

Essential 1: Gemini with your UM account

1

Go to gemini.google.com.

2

Sign in with your **UM IT account**, not your personal Gmail.

3

Check the account picture in the corner shows your UM account.

INCLUDED WITH UM

- Latest Gemini models
- Deep Research
- Gems (your own custom assistants)
- Canvas
- Audio Overviews and Gemini Live
- Gemini Notebook (NotebookLM)

UM IT Services. Usage limits may apply to some features.

Essential 2: a free ChatGPT account

1

Go to chatgpt.com and choose **Sign up**.

2

Use an email address, or sign in with Google, Microsoft or Apple.

3

Settings → Data controls → turn off **Improve the model for everyone**.

The free plan is enough for this course. Using the email you will use for AI for All makes the upgrade easier later.

Optional: AI for All (AI Għal Kulħadd)

- A **free national AI course**, developed by the University of Malta with MDIA.
- Online and self-paced: three short modules, then a final exam.
- Hosted on Coursera, in English or Maltese.
- Launched in May 2026.

ai4all.gov.mt

Who can join, and what you get

WHO CAN JOIN

- Maltese citizens **and residents**
- Aged 14 or over
- With an active **e-ID** account

WHAT YOU GET

- A verifiable digital certificate
- **One year free** of ChatGPT Plus *or* Microsoft 365 Personal with Copilot

AI for All FAQ, ai4all.gov.mt/faqs. Conditions apply.

How to sign up

1

Go to ai4all.gov.mt and choose **Register with e-ID**.

2

Pick a language, fill in the short onboarding form and accept the terms.

3

Go to **My Catalogues** and activate the **AI Programme Bundle**.

4

Click **Access Course** and complete the three modules on Coursera.

5

Take the **final exam** on the platform.

6

Your certificate and subscription code arrive by email, and appear in **My Profile**.

Tips for AI for All

- Check that the email linked to your e-ID is one you actually read. The code is sent there.
- When you choose your reward, pick **ChatGPT Plus** if you want it for this course.
- Stuck on the platform? Use **Help & Support** on the dashboard, or email servizz@gov.mt.
- Not eligible? No problem: everything in this course works on the free plans.

Already with Revolut?

REVOLUT PREMIUM

Includes **ChatGPT Go for 6 months**.

REVOLUT METAL

Includes **ChatGPT Go for 12 months**.

ChatGPT Go sits between Free and Plus: higher limits than Free. Not a sponsor, and not worth signing up for just this. But if you already pay, claim it.

revolut.com/en-MT, plan pages, October 2026. It replaced Perplexity Pro in September 2026.

Optional: try Claude

- Free at claude.ai.
- Good for comparing answers. We will not need it for the exercises.
- If you sign up, turn off model improvement in Settings → Privacy.

BEFORE NEXT WEDNESDAY

Your setup checklist

1
Open Gemini with your **UM account**.

2
Create a free **ChatGPT** account and turn off training.

3
Send the prompt on the next slide to **both**, and save the answers.

4
Optional: start **AI for All**.

5
Optional: Revolut users, claim ChatGPT Go.

6
Stuck? Bring your laptop early next week and we will fix it together.

HOMEWORK PROMPT

Your first real prompt

```
I am studying [your course] at the University of Malta.
Suggest three ways AI could help me with my studies this semester,
and one way it could cause me problems.
```

Compare the two answers. Which was more useful, and why? Bring one observation next week.

Today in five points

1

AI chatbots predict what comes next, from patterns learned on huge amounts of text.

2

Likely is not the same as true: check facts, numbers and sources.

3

The big three are ChatGPT, Gemini and Claude. Compare their answers.

4

Do not paste other people's personal data. Use your UM account for uni work.

5

Set up Gemini (UM) and ChatGPT this week.

6

You are the manager: AI drafts, you decide.

NEXT WEEK · 4 NOVEMBER

Tokens and better results

How AI reads your words, why long chats go wrong, why AI is expensive, and how to ask so you get what you need.

Questions?